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S. Gross and N. Zumbulyadis

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Use of high mixing frequencies for rf generation and detection in a multinuclear and cross-polarization magic-angle-spinning NMR spectrometer

S. Gross and N. Zumbulyadis

Research Laboratories, Eastman Kodak Company, Rochester, New York 14650

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Use of intermediate frequencies (IFs) above 100 MHz (rather than lower mixing frequencies) in a high-field multinuclear NMR spectrometer simplifies rf generation and detection, reduces rf filtering requirements, increases the frequency range of the rf synthesizers, and minimizes "spurious" responses. Simplified rf generation, gating, phase shifting, and detection electronics using 120- and 130-MHz IF's are described for a 4.7-T (200 MHz ^1H) NMR spectrometer designed for high-resolution, solid-state, and multinuclear experiments. The circuitry can be readily modified to function with the highest field (500 or 600 MHz ^1H) solenoids available.

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INTRODUCTION

This paper presents our approach to the design and implementation of simplified rf electronics for a high-field pulsed spectrometer used for multinuclear or cross-polarization magic-angle-spinning (CP/MAS) NMR. The observe and decoupler transmitters should be able to produce accurate 0° , 90° , 180° , and 270° phase shifts and be quickly settable to any Larmor frequency. The quadrature receiver should have the highest possible sensitivity, be free from most "spurious" and image responses, and be immune to overload or interference from the high-power decoupling used in CP/MAS NMR. These requirements are most easily satisfied when intermediate frequencies (IFs) higher than 100 MHz are used for rf generation, phase shifting, and phase detection. We discuss the advantages of this method over rf processing by either direct generation and detection or by frequency changing with lower (2–30 MHz) IFs and its application to very high-field spectrometers. A multinuclear and CP/MAS spectrometer designed and assembled in our laboratory and the modules for phase shifting, gating, rf generation, and quadrature detection, which use 120- and 130-MHz IFs, are described.

I. THE "HIGH IF" APPROACH

Many papers have described the construction of new multinuclear instruments^{1–8} or the "broadbanding" of older, dedicated-frequency spectrometers.^{9–11} The simplest method is to generate and receive the Larmor frequencies directly with the aid of commercially available frequency synthesizers, wideband amplifiers, gates, and phase detectors.^{1,5,7} This method has serious drawbacks. Although a broadband receiver can have $<1\ \mu\text{s}$ recovery times^{5,12} and is, therefore, highly desirable for certain solid-state experiments, its high, unselective amplification makes it susceptible to overload from the proton decoupler unless extensive rf filtering and shielding are

used. Broadband transmitter phase shifters and receiver quadrature detectors that operate directly at the Larmor frequency⁵ require wide-band quadrature hybrids that normally work over a maximum frequency range of one decade,¹³ with amplitude and phase variations. The on/off ratio in wideband transmitter rf gating circuits deteriorates with increasing frequency, and consequent rf carrier and noise leakage can degrade receiver performance. rf generation and detection via a low (2–30 MHz) fixed intermediate frequency^{2,3,8} can alleviate these shortcomings.

Use of a relatively high IF ($>100\ \text{MHz}$) has additional advantages, including simplification of rf generation and reception by reducing the filtering requirements necessary to remove image frequencies, extension of the frequency range of the rf synthesizer, and reduction of receiver "spurious" responses. To appreciate how a high IF leads to these benefits, it is useful to review the way in which signals are generated by mixing. A typical double-balanced mixer¹⁴ with IF I fed to its "L" port and synthesizer rf S applied to its "R" port produces a spectrum of output frequencies¹⁵ F at the "I"

$$F = |n_1 S \pm n_2 I|; \quad n_1, n_2 = 0, 1, 2, 3, \dots \quad (1)$$

port. For each pair of nonzero n_1 and n_2 , two output frequencies are generated. For $n_1 = n_2 = 1$, one value of F is the desired "real" signal and the other is the unwanted "image." Cases where n_1 and/or $n_2 \neq 1$ have lower intensity are discussed in Sec. IB.

A. Frequency generation and detection at high IF

Consider a double-balanced mixer used over a frequency range of $\nu^{103}\text{Rh} = 6.29\ \text{MHz}$ to $\nu^{205}\text{Tl} = 115.42\ \text{MHz}$ (at $H_0 = 4.7\ \text{T}$, $\nu^1\text{H} = 200\ \text{MHz}$). If $F = S - I$ and $I = 30\ \text{MHz}$, images will range from 66.29 to 175.42 MHz. If I is raised to 130 MHz, synthesizer frequencies of 136.29–245.42 MHz will be required, and images will

occur from 266.29 to 375.42 MHz. Note that at the higher IF the entire band of images is widely separated from the desired rf band. Therefore, a single fixed-tuned, 0–120 MHz low-pass filter can be used to pass the desired signals and reject all images.

Mixing is bidirectional and may be reversed for signal reception. A portion of the synthesizer output is mixed with the preamplified probe signal F , and IF signals are generated for

$$F = |m_1 S \pm I|/m_2; \quad m_1 = 0, 1, 2, 3, \dots \quad (2)$$

$$m_2 = 1, 2, 3, \dots$$

For the desired case where $m_1 = m_2 = 1$, the image signal must be removed or the signal/noise ratio will be degraded by as much as $2^{1/2}$.¹⁶ This can be done for $I = 130$ MHz simply by inserting a lowpass filter (identical to the one used for rf generation) between the preamplifier and mixer. In summary, if $I = 130$ MHz all Larmor frequencies up to $\nu^{205}\text{Tl}$ may be generated and received by a bidirectional mixing process requiring only two identical 120-MHz lowpass filters. This scheme permits a single broadband (untuned) preamplifier to be used, without degrading the signal/noise ratio. Furthermore, the receiver lowpass filter will also reject the 200-MHz decoupler signal.

In the above discussion a high synthesizer frequency of 245.42 MHz was used with a 130-MHz IF to generate the 115.42-MHz ^{205}Tl frequency. Note that setting $S = |I - F| = 14.58$ MHz is also acceptable [with restrictions described in Sec. I(B)]. Furthermore, ^{19}F to ^3H frequencies of 188.16–213.33 MHz can also be generated and received with $S = 58.16$ –83.33 MHz. The images will fall between 71.84 and 46.67 MHz and can be easily removed with a 186–216-MHz bandpass filter. Frequencies up to 290 MHz can thus be generated and received using only a 0.1–160-MHz synthesizer and 130-MHz IF.

B. Spurious responses

There is nothing inevitable about “spurious” responses. Their causes are outlined in Secs. I(B) and I(C); they can be minimized if a specific high IF is chosen on the basis of the magnetic field strength to be used.

A spurious response is a receiver output for some input frequency F in Eq. (2) that arises when m_1 and/or $m_2 \neq 1$. Relative intensities at these frequencies are functions of the signal input and synthesizer drive levels applied to the double-balanced mixer used for IF conversion in the receiver and of the characteristics of the mixer itself. We have observed the strongest responses when m_1 is a small odd integer and $m_2 = 1$. Equation (2) shows that the spurious responses become more widely separated and occur at higher frequencies as I and S are increased. The probability is reduced that they will fall within the passbands of the lowpass or bandpass filters discussed here. Therefore, raising I (and S) reduces spurious responses.

Particularly strong “internally generated” receiver re-

sponses also occur when

$$S = I/n; \quad n = 1, 2, 3, \dots \quad (3)$$

corresponding to frequencies

$$F = (n + 1)I/n. \quad (4)$$

Fortunately, the “forbidden” synthesizer frequencies are avoidable simply by choosing $S > I$ in Eq. (2).

The above argument also applies to frequency *generation*, as is apparent from Eq. (1). Transmitter signals generated by the wideband procedures described earlier will have improved spectral purity as the IF is raised.

C. IF selection

A suitable IF should be chosen on the basis of ease of generation, highest synthesizer frequency available, and proton resonance frequency $\nu^1\text{H}$, i.e., the field strength to be used. The IF may be a simple integral multiple of a readily available, phase-locked reference signal, such as 10.00 MHz. The best possible case is to select $I \geq 1.2 \nu^1\text{H}$ and $S > I > F$, where F is any Larmor frequency. This frequency hierarchy minimizes spurious responses and is the basis for high-quality “up-conversion” superheterodyne receivers.¹⁷ Santini and Grutzner⁶ have advocated this type of receiver in their design of a multi-nuclear NMR spectrometer. Lowpass and bandpass filters covering 0–0.6 $\nu^1\text{H}$ and 0.9–1.1 $\nu^1\text{H}$ (0.7–1.1 $\nu^1\text{H}$ to include ^3He) will remove image signals for any nuclide. This filter combination will also suppress the proton decoupler signal and any decoupler amplifier noise at the Larmor frequency at the receiver input. If a superconducting magnet is used and it is not possible to satisfy $S > I > F$, we recommend lowering the IF to near 0.65 $\nu^1\text{H}$. The synthesizer frequency should still be $> \text{IF}$ if possible. The electronics described in Sec. II are based on this compromise.

Equations (2) and (4) should be used to check for possible spurious responses or “forbidden” synthesizer frequencies. If conflicts occur, the IF may be lowered further into the passband of the lowpass filter. This is permissible if the IF does not fall within the spectral range of any observable nuclide and the receiver IF rejection is good enough [$m_1 = 0$; $m_2 = 1$ in Eq. (2)]. For example, a 140-MHz IF and 500-MHz frequency synthesizer such as the PTS 500¹⁸ should be appropriate for “500 MHz” and future “600 MHz” magnets. Note that the lowpass and bandpass filters described earlier can be used with any of the IFs suggested (except $\nu^{205}\text{Tl}$ at $^1\text{H} = 600$ MHz).

II. SPECTROMETER DESIGN

Our NMR spectrometer was designed for multinuclear and ^{13}C CP/MAS studies and was built around an Oxford Instruments¹⁹ 4.7-T (200-MHz proton) 98-mm bore magnet. This field strength was chosen to be as high as possible to maximize multinuclear sensitivity and still be low enough for solid-state magic-angle-spinning experiments with the spinning speeds obtainable from current

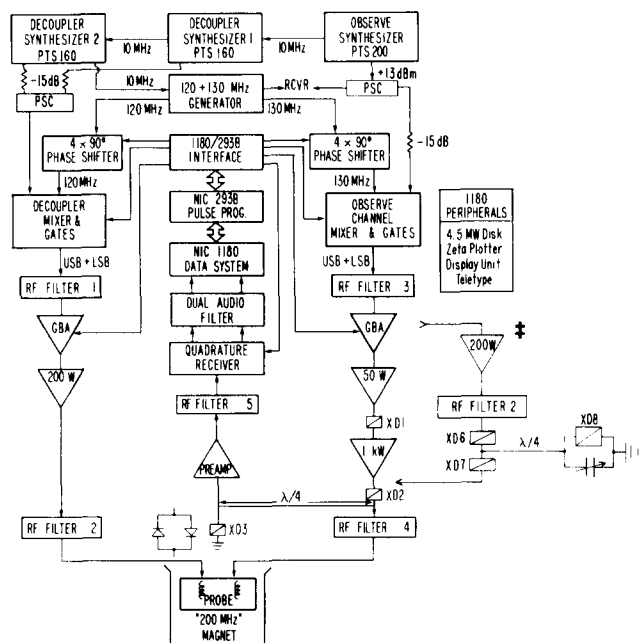


FIG. 1. Block diagram of the entire spectrometer. PSC = power splitter/combiner. GBA = gated bilevel amplifier. XD1-8 = crossed diodes. ‡188-213-MHz configuration; see text.

CP/MAS probe technology. Commercially available components were used if they met our design requirements. Shim coils and variable-temperature multinuclear probes were furnished by Cryomagnet Systems.²⁰ The probes cover ranges of 5-10, 10-20, and 20-40 MHz with 25-mm tubes, 40-80 MHz with 20-mm tubes, and 188-213 MHz with 5-mm tubes. The solid-state ^{13}C probe was supplied by Chemagnetics²¹ and uses "bullet" type rotors that spin routinely at >4 kHz. The spectrometer is controlled by a Nicolet²² data system with a 293B pulse programmer. Specialized rf and control electronics constructed in-house include an IF reference generator; two gated, double-sideband rf generators and phase shifters; two gated, bilevel driver amplifiers; a quadrature receiver; and an NMR interface. These modules are described in detail below.

Figure 1 is a functional block diagram of the spectrometer. Observe and decoupler electronics are both multinuclear and have almost identical broadband circuitry. Observe frequencies are generated by a PTS 200 synthesizer. Two PTS 160 synthesizers are used for double and triple resonance decoupling and are phase locked to the PTS 200 synthesizer by cascading its 10-MHz reference. The observe and decoupler IFs are 130 and 120 MHz, respectively. Different IFs are used to preclude receiver interference from decoupler modulation or gating. They are produced by a 120- and 130-MHz generator supplied with a 10-MHz reference from the second PTS 160 synthesizer. All spectrometer frequencies are thereby phase locked. Portions of the PTS 200 synthesizer and the 130-MHz output are also used by the quadrature receiver.

Frequency generation starts by phase shifting the IFs by 0° , 90° , 180° , or 270° . The synthesizer frequency S is mixed with the result, and upper and lower sidebands

(USB + LSB) are generated. Either sideband could be the desired output. For the observe channel, $S = 130 + F$ for $F < 70$ MHz, and $S = |130 - F|$ for higher frequencies. For example, $S = 136.29$, 49.02 , or 70.00 MHz for ^{103}Rh , ^{31}P , and ^1H Larmor frequencies of 6.29, 80.98, or 200.00 MHz, respectively. The resulting images of 266.29, 179.02, or 60.00 MHz are widely separated from the desired frequencies and are easily removed by rf filter 3. This is either a Cir-Q-Tel²³ FLT/20-120-15/50-3A/3A 120-MHz lowpass or an FBT/20-201/30-5/50-3A/3A 186-216-MHz bandpass unit. rf filter 1 and decoupler frequency generation are identical, except that $S = |120 \pm F|$. rf gating also occurs at this time.

Identical computer-controlled, gated, bilevel driver amplifiers drive the power amplifiers at either of two preselected output levels. Output from a 50-W ENI²⁴ 350L 0.12-120-MHz power amplifier drives the 1-kW Henry Radio²⁵ Tempo 2006 50-54-MHz amplifier at $\nu^{13}\text{C} = 50.3$ MHz for CP/MAS experiments. The ENI amplifier is used alone for liquid-state multinuclear NMR. An Amplifier Research²⁶ 100 LM9 broadband amplifier rated at 100 W at 0.01-220 MHz furnishes decoupling power. (Our unit supplies >230 W at 1-200 MHz.) rf filter 2, a high-power (300 W) version of the Cir-Q-Tel 186-216-MHz bandpass filter described above, is used to attenuate any internally generated 100LM9 amplifier noise at the observe frequency. The 100LM9 amplifier is also the observe channel transmitter for ^{19}F , ^1H , and ^3H .

Crossed diodes XD2 and XD3 and the quarter-wave ($\lambda/4$) line form the popular transmit/receive duplexer of Lowe and Tarr.²⁷ The 188-213-MHz configuration is added at the highest frequencies and gives additional suppression of background noise generated in the AR power amplifier. Crossed diodes consist of three or more pairs of low-capacitance 1N3062 or Sylvania²⁸ ECG519 diodes in series. XD2 is replaced by four banks of eight pairs of 1N914B crossed diodes when the 1-kW ^{13}C amplifier is used. XD3 is one pair of 1N3062 diodes mounted with the shortest leads, as close to the preamplifier as possible. Doubly-shielded, low-leakage RG223/U cable and Berk-Tek²⁹ miniature, flexible RG8X line are used for all low- and high-power rf interconnections, respectively. The two $\lambda/4$ lines are the only "tuned" elements in the spectrometer besides the probes.

A Watkins-Johnson³⁰ CA71 5-200-MHz broadband preamplifier with a 2.1-dB noise figure and 17-dB gain amplifies all nuclear frequencies. rf filter 5 removes noise

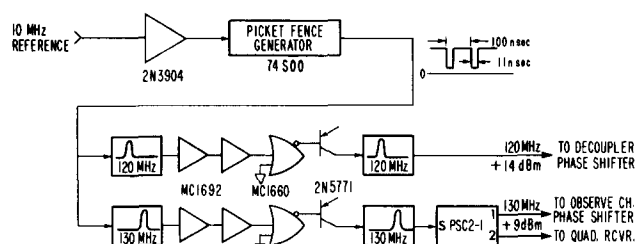


FIG. 2. Block diagram of the 120- and 130-MHz reference generator.

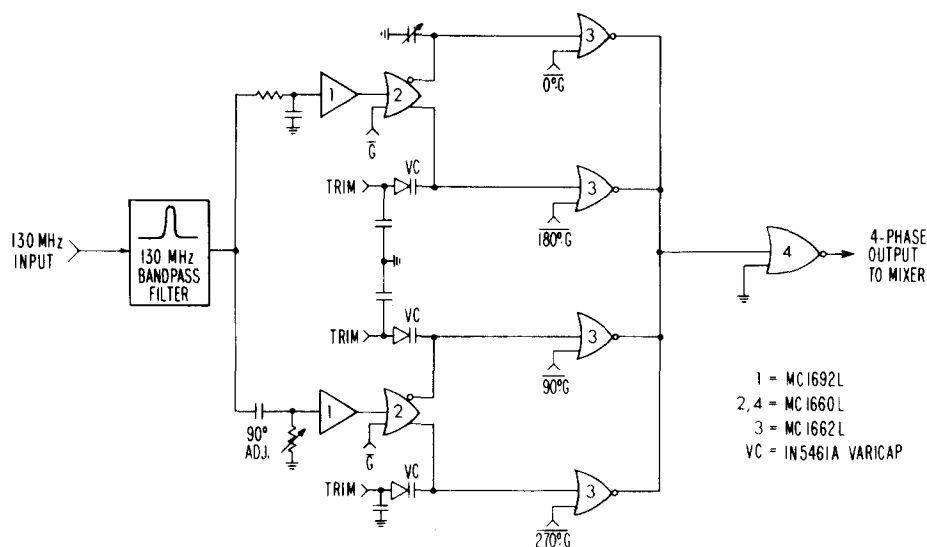


FIG. 3. Block diagram of the ECL phase shifter; 470- Ω pulldown resistors not shown.

at the image frequency originating either in the preamplifier or receiver coil. We use a Cir-Q-Tel FLT/20-100-23/50-3A/3A 100-MHz lowpass unit, although one of the types used for rf filter 3 would probably suffice. rf filter 4 [Cir-Q-Tel HPL/25-100-23/50-1A(S)/1A(S)] is a high-power version of filter 5 that can handle the ^{13}C power used in CP/MAS experiments. It is used to attenuate the ^1H decoupler frequency leakage from the probe, thereby preventing preamplifier or receiver overload. This filter has only 0.2-dB attenuation at 50 MHz and has a negligible effect on the signal/noise ratio. rf filter 4 is removed when nuclei with frequencies above 80.98 MHz ($\nu^{31}\text{P}$) are observed.

Quadrature receiver outputs are conditioned by two Frequency Devices³¹ FC 103/1740BT-4 4-pole lowpass Butterworth filters. They are adjustable in 100-Hz steps from 0.1 to 50.0 kHz with three BCD thumbwheel switches but can also accept and store strobed TTL data if computer control is desired. Amplitude and phase matching are enough to keep images in quadrature NMR spectra below 0.5%.

A. 120- and 130-MHz generator

The 120- and 130-MHz mixing frequencies are derived from 10-MHz harmonics of a picket-fence generator driven from the PTS synthesizer reference (Fig. 2). Hoult and Richards⁷ used this approach to extend by a factor of 10 the upper range of a synthesizer in their supercon NMR spectrometer. We generate picket-fence pulses by NANDing Q and delayed $\bar{Q}$ outputs of an R - S flip flop driven by normal and inverted 10-MHz signals. Harmonics at 120 and 130 MHz are selected by simple L - C filters. MECL³² 1692 and 1660 ICs are used as linear rf amplifiers and limiters, respectively. A Mini-Circuits PSC2-1 power splitter/combiner provides isolated 130-MHz outputs for the observe-channel phase shifter and quadrature receiver. Additional circuits (not shown) generate several buffered 10- and 1.0-MHz TTL outputs for future use. The complete circuit is assembled on one 11.4×16.5 cm Vector³³ 3662A-6 ground-plane card and is contained in a Vector HFE 20-46-97 rf shielded case.

The same ground-plane cards and cases are also used for assembly of the double-sideband rf generators and bilevel amplifiers described below. Single $n \times 10$ -MHz outputs covering 10–140 MHz recently became available on PTS 160 and 200 synthesizers.

B. Gated double-sideband rf generators

These modules accept IF, synthesizer, and logic inputs and generate phase-shifted, gated, upper and lower sidebands. Each unit contains a power splitter/combiner, $4 \times 90^\circ$ phase shifter, mixer, and gates (Fig. 1). A decoder and four pulse stretchers drive four panel-mounted light-emitting diodes (LEDs) that indicate phase and output status.

Phase shifting is done at the fixed IF and is translated to the desired frequency by the mixing process. This procedure ensures that phase accuracy is independent of the output frequency. We used high-speed MECL III integrated circuits (Fig. 3). The IF is filtered by a simple L - C circuit and then split into two signals 90° apart by the R - C network. Each signal is amplified and converted to ECL logic levels by chip 1. Chip 2 generates four gated rf phases at its two pairs of complementary outputs. Three panel-mounted trimpots adjust the four phases to exact 90° differences by varying the bias voltages on three voltage-variable capacitance diodes. Chip 3 selects the desired phase shift, while chip 4 accepts the wire-OR result and functions as a limiter to provide a constant output level.

This circuit requires less room and about a third of the number of parts of traditional passive phase shifters,⁵ because the *ECL ICs act as rf amplifiers, 180° power splitters, rf gates, power combiners, and limiters*. This circuit also simplifies system design because it works at the actual IF rather than a higher multiple, as done in some other digital phase-shifting schemes^{34,35} including those using ECL circuits.³⁶

The rf mixing and gating scheme is shown in Fig. 4. Both synthesizer and phase-shifted IF outputs are gated before mixing for a high on/off ratio.⁹ The synthesizer

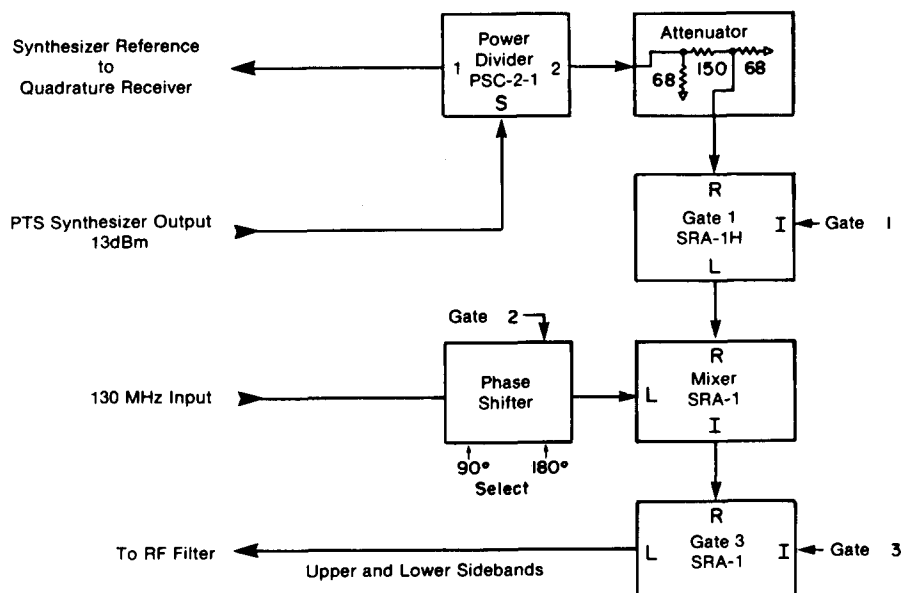


FIG. 4. Block diagram of the gated double-sideband rf generator.

signal is gated with a Mini-Circuits double-balanced mixer, with a driver circuit similar to that of Karlicek and Lowe.⁵ The phase shifter is inherently gated. The double-sideband mixer output is also gated, although this may not be necessary. Small transients generated by gates 1 and 3 are minimized by proper choice of the current level applied to the *I* ports. The synthesizer must not see the impedance change at gate 1 when it is turned on or amplitude will fluctuate undesirably at the start of the rf pulse. This transient is minimized by using the maximum (+13 dBm) synthesizer output and by isolating gate 1 from the synthesizer with the power divider and attenuator.

C. Gated bilevel driver amplifiers

The observe and decoupler broadband bilevel driver amplifiers (Fig. 5) are identical. Each contains a Watkins-Johnson CA76 5–500-MHz amplifier with 28-dB

gain and +15-dBm power output. This is gated by a Mini-Circuits 5–500-MHz RAY-1 high-level mixer to suppress CA76 amplifier noise when the rf is off. Crossed diodes could have been used instead, but they would have distorted the output waveform. A Mini-Circuits PAS-1 5–450-MHz current-controlled attenuator is driven by one of two TTL-selectable, 10-turn potentiometers mounted on the front panel. Two adjacent LEDs simultaneously indicate which potentiometer is “active” and the presence of rf output. Each bilevel amplifier has 21-dB gain and +3.2-dBm output at zero attenuation, which is enough to drive the ENI or AR power amplifiers into saturation. The maximum attenuation range is 47 dB, although an undesirable gain drift of 1–2 dB occurs near maximum attenuation if the output is gated on for more than a few minutes.

The bilevel amplifier and double-sideband rf generator are gated on and off at the same time. This simplifies the NMR interface design (see below) and is permissible,

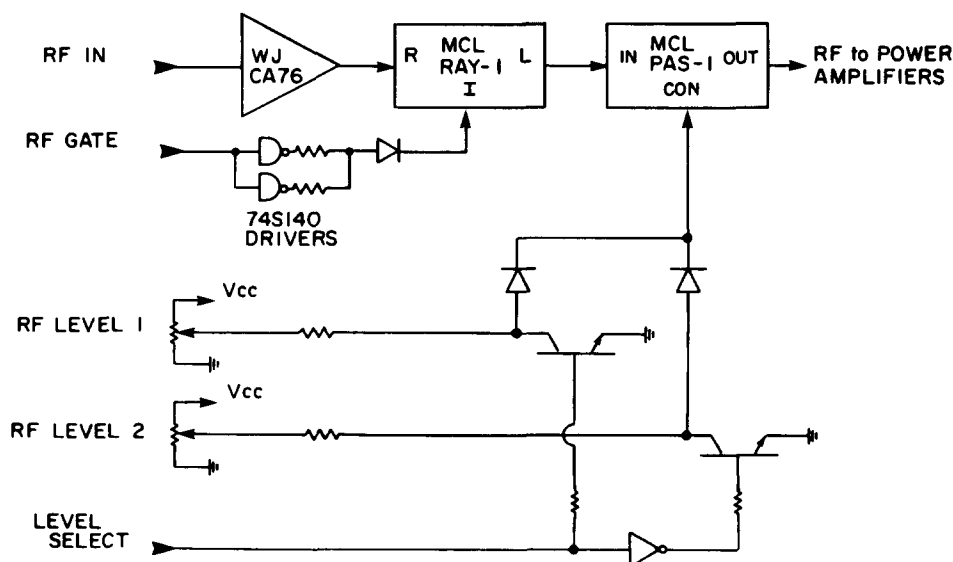


FIG. 5. Block diagram of the gated bilevel driver amplifiers.

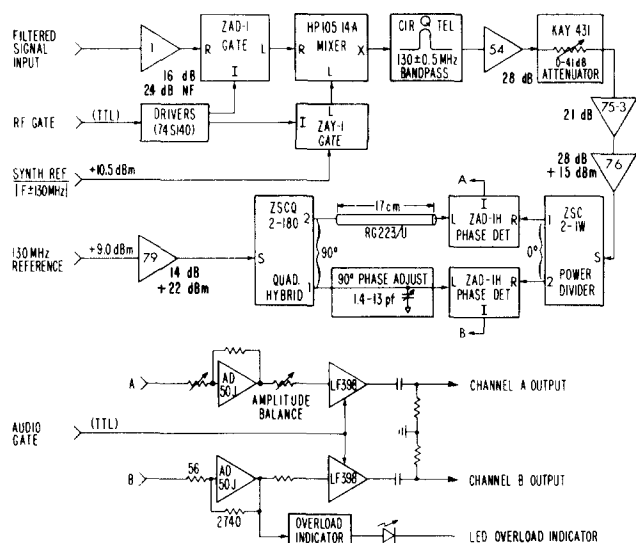


FIG. 6. Block diagram of the quadrature receiver.

because lowpass filter 3 and bandpass filter 1 (Fig. 1) introduce negligible (and accountable) delays of only 18 and 50 ns, respectively.

D. Quadrature receiver

The quadrature receiver is constructed largely with commercially available components and operates at an IF of 130 MHz. Broadband Watkins-Johnson CA series amplifiers are used throughout. These units come in fully shielded $21 \times 12.5 \times 25$ cm cases with SMA connectors and completely eliminate any discrete rf wiring.

A block diagram is shown in Fig. 6. A Hewlett-Packard³⁷ HP 10514-A double-balanced mixer converts the input signal to 130 MHz. A Mini-Circuits ZAD-1 mixer would probably perform equally well. Both the input and synthesizer signals are gated before mixing to improve the on/off ratio.⁹ A Mini-Circuits ZAY-1 very high-level mixer is needed to gate the 10.5-dBm synthesizer level. Overload resistance to extraneous signals is maximized by keeping the gain as low as possible before mixing but not low enough to degrade the receiver noise figure. A Cir-Q-Tel (S)FBT/20-130/1.5-5/50-3C/3D bandpass filter with controlled phase characteristics gives an IF passband of 130 ± 0.5 MHz. The bandwidth can be changed easily by substituting a different filter. The three Watkins-Johnson amplifiers used for IF amplification provide necessary gain and will not oscillate at any setting of the Kay³⁸ 0-41-dB attenuator. Two Mini-Circuits high-level mixers are used as 130-MHz phase detectors and offer improved dynamic range and lower IM distortion than the standard-level ZAD-1 types. The required 17-dBm 0° and 90° references are supplied by a Mini-Circuits ZSCQ2-180 quadrature hybrid driven by a CA79 amplifier. A 1.4-13-pF capacitor (E. F. Johnson³⁹ 193-0004-005) in a Pomona⁴⁰ 2391 box is used to trim the 90° phase difference. Two Analog Devices⁴¹ 50J operational amplifiers with 70-MHz gain bandwidth product are operated at a gain of 49 and have flat response

to >100 kHz. The bandwidth could be increased, if necessary, to about 1 MHz by cascading two 50J amplifiers, each with a gain of 7. These amplifiers clip before any other component in the receiver saturates, even with full 41-dB attenuation. Imminent overload is indicated by a panel-mounted LED. Two National Semiconductor⁴² LF398 monolithic sample and hold circuits provide audio (af) gating. Additional circuitry minimizes LF398 switching transients. The 1180/293B interface (below) opens the af gates 3 μ s after the receiver rf gates and shuts the af gates 2 μ s before turning off the rf gates. The LF398 circuits have a bandwidth >100 kHz and low-impedance outputs. Optional ac coupling with 10-s time constant eliminates any dc offset in the FIDs. Total gain of the receiver at 200 MHz is 104 dB (without the CA71 preamplifier).

E. 1180/293B interface

The 1180/293B interface establishes communication among the Nicolet 1180 data system, the 293B pulse programmer, and our rf electronics. Liberal use of data selector logic (74151, 153) facilitated the current design and will simplify future modifications. In addition to controlling observe and decoupler gating, receiver gating, phase, and power levels, the interface also generates timing sequences for homonuclear and broadband heteronuclear decoupling and contains several interlocks which may be separately disabled with panel-mounted switches. The interlocks turn off the observe and decoupler rf gates and stop data acquisition if cooling air is lost or if operator error or system malfunction causes an excessive observe or decoupler duty cycle. A simple circuit to monitor the duty cycle is shown in Fig. 7.

F. Initial adjustments and sample spectra

Initial adjustment of the rf electronics was straightforward and rapid. The four phase shifts were adjusted to 90° differences by a simple visual technique: a 293B program advanced the rf phase by 90° every millisecond. A 5-MHz output from the driver amplifier was displayed on a Tektronix⁴³ 475A oscilloscope with the bandwidth set to 20 MHz. The scope was triggered externally at 20 MHz with a second phase-locked synthesizer. Expansion of the zero crossings showed two groups of four lines each. The three trim pots on the gated double-sideband generator were then adjusted to cause coalescence of each

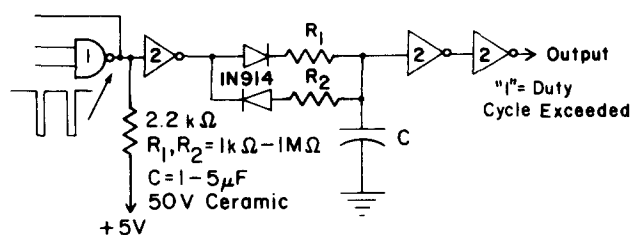


FIG. 7. Simple circuit to monitor the duty cycle of a pulse train. Duty-cycle trip point is adjustable over a large range by selection of R_1 , R_2 , and C . Output will drive one "LS" TTL load. Chip 1 is any TTL gate; chip 2 is 1/6 of a CMOS 4069UB.

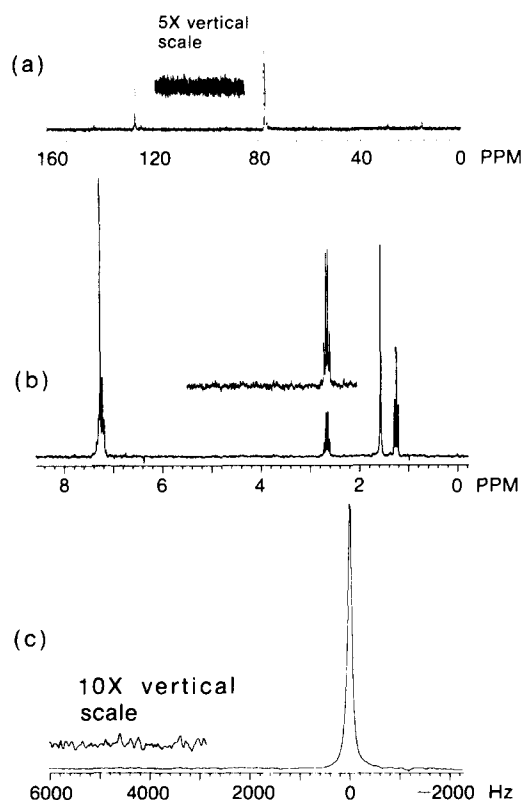


FIG. 8. Signal/noise spectra. (a) 50.294-MHz ^{13}C spectrum of 10% ethyl benzene in $\text{CDCl}_3 + \text{CHCl}_3$; S/N = 135; 20-mm tube; 1 transient; 8-W broadband ^1H decoupling; 35- μs 90° pulse; ± 5 -kHz spectral width; 32 K FID (1.63-s AT); 0.4-Hz line broadening. (b) 199.995-MHz ^1H spectrum of 0.1% ethyl benzene in CDCl_3 ; S/N = 32; 5-mm tube; 1 transient; 5.5- μs 90° pulse; ± 2 -kHz spectral width; 16K FID zero filled to 32K (2.04-s AT); 1.0-Hz line broadening. (c) 27.112-MHz ^{17}O spectrum of neat H_2O ; S/N = 401; 25-mm tube; 100 transients; 5-W broadband ^1H decoupling; 72- μs 90° pulse; ± 12.2 -kHz spectral width; 8K FID (0.168-s AT); 1.168-s cycle time; 50-Hz line broadening.

group of four lines. Measurement of these phase shifts several months later by a new technique⁴⁴ accurate to $\pm 0.05^\circ$ showed that the previous calibrations were accurate to $\pm 0.7^\circ$ and drifted less than $\pm 0.2^\circ$ over 17 h and less than $\pm 0.3^\circ$ over 2 weeks. Amplitude balance among the four phases is within 0.032 dB.

rf switching times were measured by phase detecting the bilevel driver amplifier output at 81 MHz and observing the rise and fall times on the Tektronix 475A scope with the bandwidth set to 20 MHz. Phase shifting occurs in 50 ns. Pulse rise time is 0.5 μs to 90% and 2 μs to 100% of full amplitude and is accompanied by a decaying transient at the synthesizer frequency, which disappears after 5–10 μs . We believe these effects at pulse turn-on time are caused by dc coupling between the output of the 130-MHz phase shifter and the double-balanced mixer and are correctable by using ac coupling. Pulse fall time is 15 ns. The bilevel driver amplifier outputs take about 25 μs to stabilize after switching levels.

This time should be reducible to $< 0.1 \mu\text{s}$ if a RAY-1 or SRA-1H mixer is substituted for the PAS-1 attenuator in this module, if some distortion of the output signal is acceptable. We have not observed any degradation in multinuclear or CP/MAS results due to any of the above switching times.

Spectral purity at the output of the bilevel driver amplifiers was measured with a Hewlett-Packard⁴⁵ spectrum analyzer consisting of an 8443A tracking generator/counter, 8552B and 8554B IF amplifiers, and 141T display. The strongest "spurious" signal at 5–100 MHz was 36 dB below the desired output and 32 dB down at 188–213 MHz. Sidebands at ± 10 MHz from the output frequency were minimized by fine tuning the 120- and 130-MHz generator and were 39 and 46 dB down in the decoupler and observe channels. No leakage of the desired rf signal was observed when the rf gates were off.

Amplitude and 90° phase balance between the two channels in the quadrature receiver was easily adjusted by applying a test signal offset 1 kHz from the carrier and calculating α and γ from the FID by the procedure of Parks and Johannesen.⁴⁶ These calculations were implemented in the Nicolet NTCFTB-1180 software as the "QC" command. Values of 0.0 and 1.0 for α and γ , respectively, indicated optimum balance.

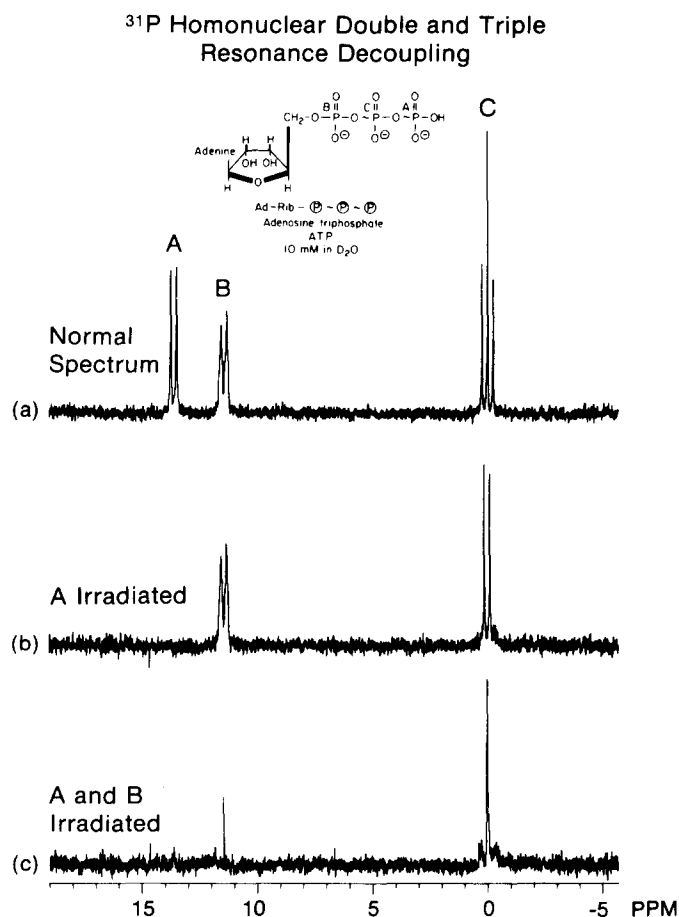


FIG. 9. 80.96-MHz ^{31}P spectra of ATP illustrating homonuclear double and triple resonance decoupling. 20-mm tube; ± 1 -kHz spectral width; 8K FID (2.04-s AT); 0.5-Hz line broadening. (a) 8 transients; 12.04-s cycle time. (b) and (c) 64 transients; 2.04-s cycle time.

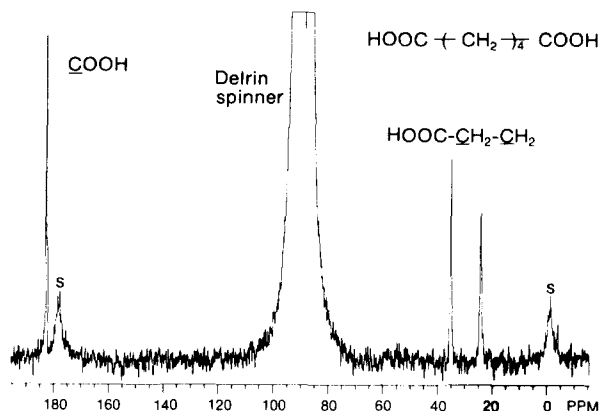


FIG. 10. 50.29-MHz ^{13}C solid-state CP/MAS spectrum of adipic acid. 4.5-kHz spinning rate; 292 transients; 115-W (12 G) ^1H decoupling; 5- μs ^1H 90° pulse; 500-W, 1-ms ^{13}C mixing pulse; ± 10 -kHz spectral width; 4 K FID (0.102-s AT) 20.102-s cycle time; 5.0-Hz line broadening. s = spinning sideband.

Spurious responses were noticed only when ^{31}P was observed at 81 MHz. If a synthesizer frequency of 49 MHz was used, the receiver was susceptible to 17-MHz interference [$m_1 = 3$ in Eq. (2)] from the 1180 computer. We used a separate 81-MHz bandpass filter to remove this signal. Generation of 81 MHz by mixing with a 211-MHz synthesizer frequency (from a PTS 500) would also eliminate this response.

The receiver IF rejection is 26 dB (0.05), thus making it feasible to place the IF within the lowpass-filter passband. This would be done (as discussed earlier) if the spectrometer were to be used with a very high-field magnet. If no interfering signal is present at the IF, the receiver noise will be increased by only $(1.0^2 + 0.05^2)^{1/2} = 1.0012$,¹⁶ which is negligible.

Receiver recovery times at 50 MHz, with a 50- Ω load substituted for the probe, were 9–17 μs , depending upon transmitter power and pulse width. These times are adequate for multinuclear and CP/MAS experiments. Hoult's⁴⁷ T/R switch might decrease this time by reducing the rf pulse voltage applied to the preamplifier. Passive voltage limiters would also help greatly, if a 2.2-dB degradation of the spectrometer noise figure is acceptable.¹²

Typical spectral results are shown in Figs. 8–10. Total drift was 0.02 ppm over 10 h, and all spectra were obtained without field/frequency lock. Quadrature detection was used in all cases and required that the frequency axis be reversed (Nicolet “SR” command) whenever ($S - I$) was negative, as for ^{31}P and ^1H spectra. Signal/noise ratios in Fig. 8 were calculated with the Nicolet “SN” command. Figure 9 illustrates multinuclear homonuclear double and triple resonance. This is accomplished by feeding the observe and decoupler bilevel driver amplifier outputs to the ENI power amplifier through a Mini-Circuits ZSC2-1W power splitter/combiner. A CP/MAS spectrum of adipic acid is shown in Fig. 10. High decoupler power does not cause receiver interference in CP/MAS experiments.

III. DISCUSSION

Advantages and simplifications in NMR spectrometer design are gained by use of high intermediate frequencies for rf generation and detection. Now that high-frequency broadband components are commonplace, we believe that this approach will be used in the construction of new NMR spectrometers, especially those with very high-field magnets. A multinuclear and CP/MAS spectrometer demonstrated the feasibility of our approach. No leakages, drifts, or other problems associated with the high IFs have been encountered in this system, which has now been in operation for about a year.

We have stressed ideas rather than specific circuit details. However, we would be happy to supply detailed schematics to anyone interested. We are also designing an automatic multinuclear field/frequency lock for our system, which will be reported later.

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